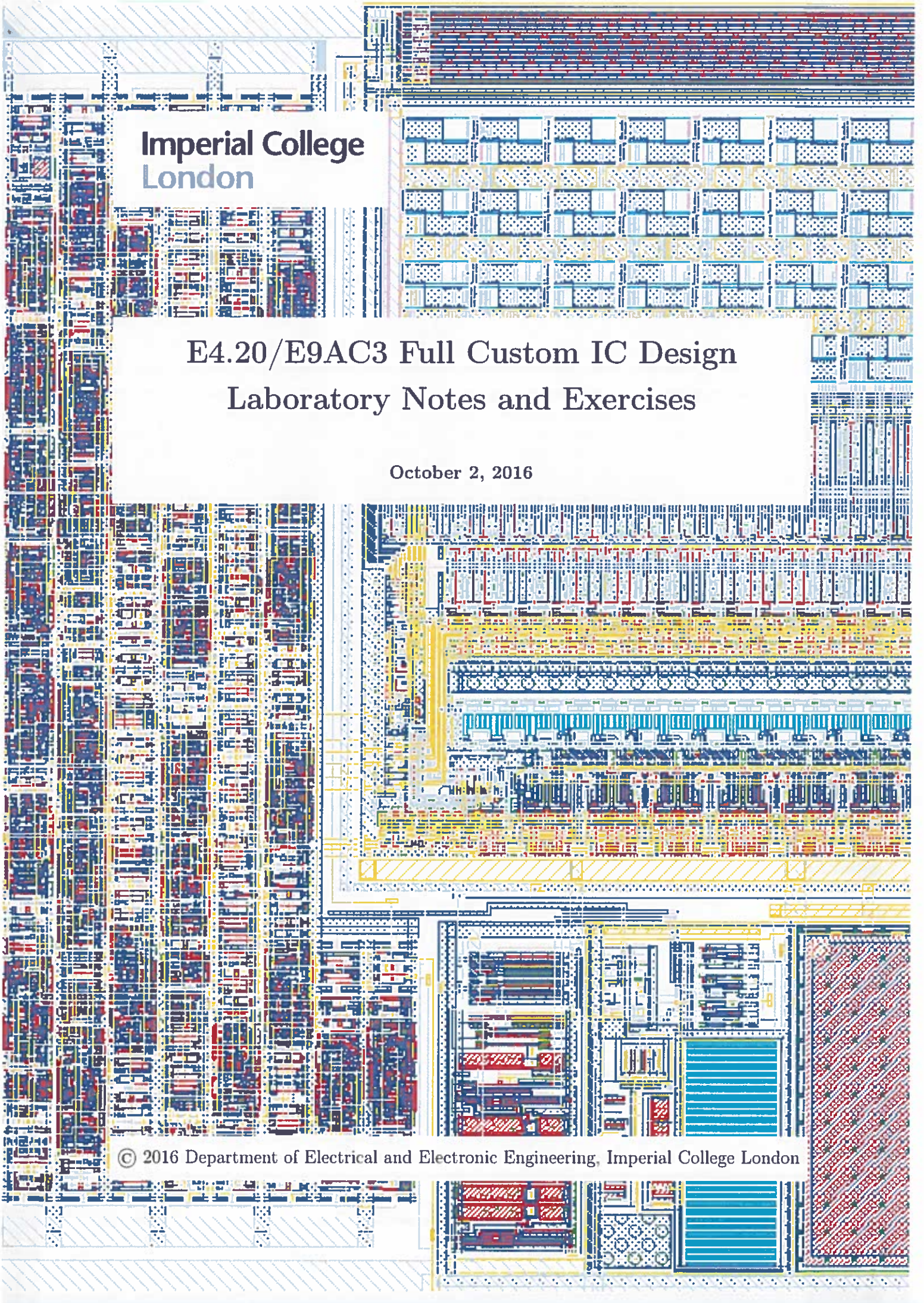


Imperial College London

E4.20/E9AC3 Full Custom IC Design Laboratory Notes and Exercises

October 2, 2016

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Introduction

1 Objective

Learn how to design full custom ICs using Cadence Design Systems, EDA Software.

2 Administration

- GTA supported Lab session: **17/10/2016, 24/10/2016, 31/10/2016, 9 AM–12 PM in room EENG 507.**
- Lab exercises are done **individually.**
- Do not **under any circumstances** copy any material from anyone else's logbook. Any evidence of plagiarism will be reported to the college in line with the academic and examination regulation 2016/17.

3 Support

- GTAs: Dr Song Luan, Dr Yan Liu, Dr Lieuwe Leene, Mr Nicolas Moser, Mr Nicholaos Miscourides, Mr Khalid Mirza, Mr Matthew Douthwaite and Mr Peilong Feng
- Questions outside lab hours should be posted via the on-line support forum on blackboard. GTAs check the forum everyday.
- Problems with computers should be reported to the computer support group.(Mr Xiangming Xiao (xiao@imperial.ac.uk)).
- Quick reference card at the end of this note
- Help button in cadence

It is highly recommended to keep all the questions on the on-line support forum to share with your colleagues.

4 Deliverables

An **electronic logbook** should be submitted via blackboard on specific dates for the two lab sessions. (See Important Dates below). This should contain:

- Schematics and any calculations made
- Design trade-offs (if any) and your understanding of a specific design

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- Results from simulations (i.e. tables and graphs) with clear labelling (axis, units, etc.)

This electronic logbook will be assessed based on the quality of documentation (clarity, readability and concise account) and interpretation of results. This should not exceed **10 A4 page limit** (minimum font size 11 and margin 2 cm) per lab. Otherwise, a penalty will be given (-20%).

The breakdown is:

- Description (30%) – Documentation clarity and readability
- Analysis (40%) – Design methodology, interpretation and analysis of results
- Layout (40%) – Layout design (compact, tidy and good practice) with evidence of clean DRC and LVS

5 Important Dates

- **27/10/2016** – logbook submission for Lab 2
- **03/11/2016** – logbook submission for Lab 3
- **07/11/2016** – Start of project lab session

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Laboratory 1 Schematic Entry and Simulation

What you will learn

- How to start Cadence IC6 integrated circuit design environment
- How to create a circuit schematic
- How to perform a DC simulation with sweeping variables and component parameters

1.1 Remote connection to EEE Cadence Server

Login to Windows and start MobaXterm from start menu. Click Session from the menu bar and setup a ssh connection as shown in Fig. 1.1. Type in your college username and make sure X11 forwarding is checked (uncheck compression if computer is on campus).

The two servers are **ee-mill1.ee.ic.ac.uk** and **ee-mill2.ee.ic.ac.uk**. You can choose either one. *Note that password won't echo back on the display while typing.*

The servers can only be connected within the college network or via VPN.

1.2 Starting a Cadence session

Type cd nfshome and press enter to change the directory to the folder called nfshome.

If you cannot find the directory or the file you stored are missing, logout and login again to get the folder loaded properly. Note that you have to run cadence within this directory or its subdirectories. **Cadence does not work properly outside "nfshome"!**

Create a directory called CadenceLab using the command mkdir CadenceLab and then use cd CadenceLab to change into that directory. **This directory will be where you start Cadence EVERY TIME.**

Type the command pdk gpdk and then cadence gpdk18 to start Cadence.[1]

Once Cadence is launched, you will see the Command Interpreter Window (CIW) as shown in Fig. 1.2. Open the Library Manger (Fig. 1.3) from Tools⇒Library Manager. Define a new library with File⇒New⇒Library called Lab1. Then in the pop-up window, select Attach to an existing technology library. This allows you to choose the technology to be attached to the library as well as its associated simulation models and layout cells. Here, select gpdk180 as shown in Fig. 1.4.

[1]The process to be used in this lab is generic .18 μm process. PDK stands for Process Design Kit.

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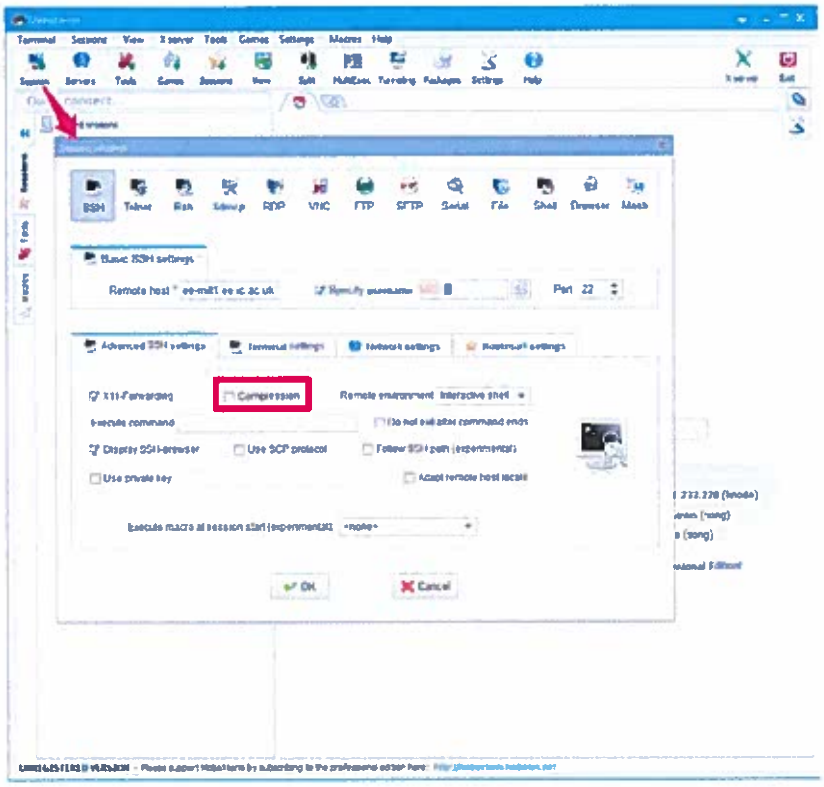


Figure 1.1: MobaXterm used for remote connection via SSH.

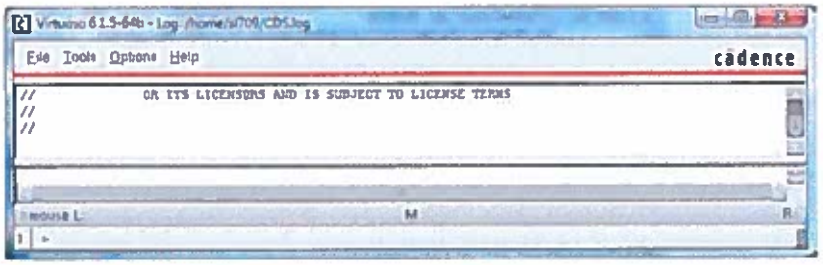


Figure 1.2: Command Interpreter Window

1.3 Create a schematic for an inverter

1.3.1 Create a cell view

From the Library Manager, click on the library you just created (i.e. Lab1) and select from the menu File⇒New⇒Cellview. Then enter the cell name "Inverter" as shown in Fig. 1.5. You may see a message asking for a license upgrade. Click "yes" or "always" to proceed.

1.3.2 Adding Components to the cell view

Components are added to the schematic by selecting instances from a library. The analogLib library contains a variety of standard components such as ground and DC voltage sources. None of the components from analogLib are linked to a specific technology (i.e. technology independent). Any component associated with the chosen technology is in gpdk180. **All MOSFETs**

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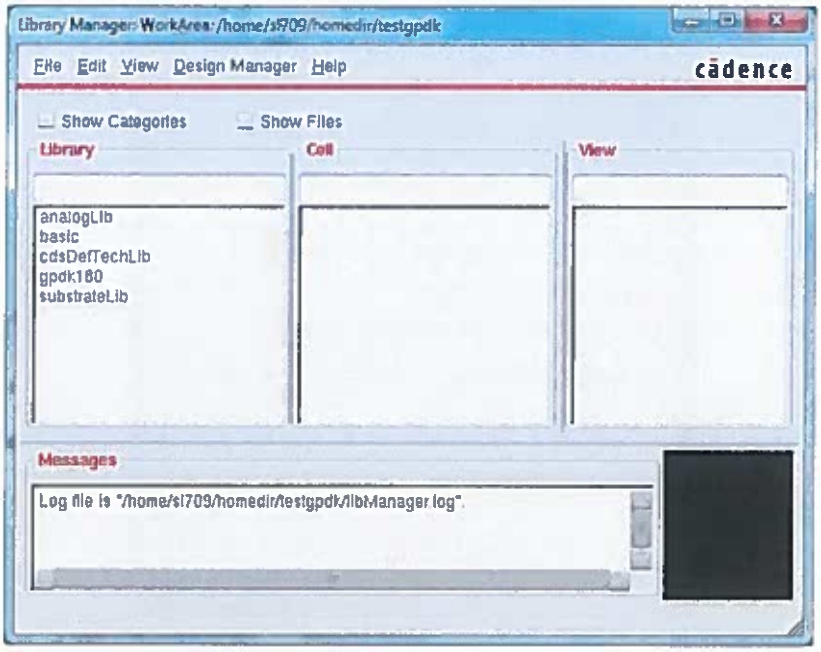


Figure 1.3: Library Manager



Figure 1.4: Select the technology library to be used

used must be taken from this library.

To place a component use the add instance button (keyboard shortcut "i") on the toolbar (Fig. 1.6), and then select Browse. Navigate through the Library Browser - Add instance (NOT the library manager!) and select the symbol view of the nmos device in the gpdk180 library. Select this and place it in the schematic. Do the same thing to add PMOS from gpdk180 library.

Now we need to place pins for the input and output using the create pin button from the toolbar (shortcut "p"). Create an input signal pin and name it "In", an output signal pin "Out", power pin "VDD" and "GND". Then use the create narrow wire button (shortcut "w") to connect all the components. The finished schematic is shown in Fig. 1.7.

1.3.3 Edit properties of a component

Click on the edit properties button from the toolbar (shortcut "q") then select the PMOS. Change its width and length to 2 μm and 180 nm respectively. Then set the width and length of the

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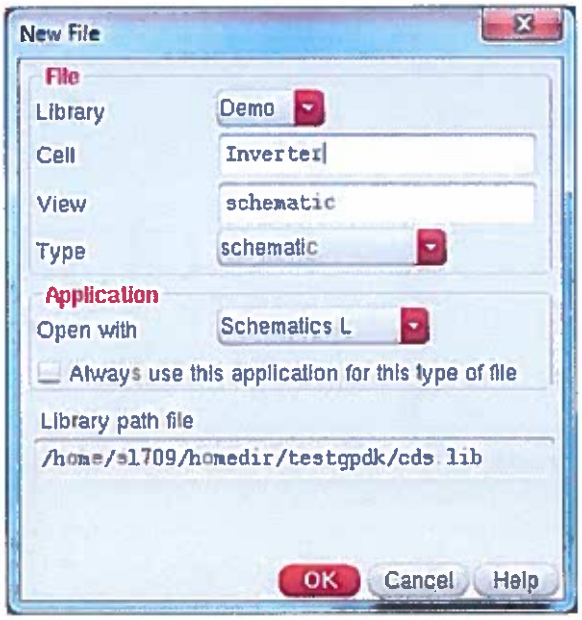


Figure 1.5: Create a new cell view



Figure 1.6: Toolbars of schematic editor

NMOS to 1 μm and 180 nm respectively. Check and save your schematic using the check and save button (shortcut "shift + x"). If nothing pops up, then it has been saved with no warnings and errors. Otherwise you should rectify errors by looking at what the CIW says and what is blinking on the schematic.

1.3.4 Create a symbol

We are going to associate the schematic we have created so far with a symbol so it can be used within a hierarchical schematic design. You should (and are strongly advised to) partition the design into separate smaller blocks. This is also important if you have many repeated instances. You should use the same symbol multiple times, rather than copy and pasting the circuit unless you want to change transistor sizes.

To create a symbol, select the menu option Create⇒Cellview⇒From Cellview.... This will open a dialogue confirming the name of the schematic from which you want to create the cell view. The default options here are fine, so click OK to proceed to the next dialogue, shown in Fig. 1.8.

The symbol generator will automatically place pins on the left, right, top or bottom of a symbol. Click on OK and the symbol editor window will open. From here you can edit the shape of the symbol. The red outline is the boundary of the symbol. The green lines are the symbol shape. The |@partName| and |@instanceName| are dynamic text labels defined when you add an instance to a schematic. You do not need to change them. The default symbol shape is a

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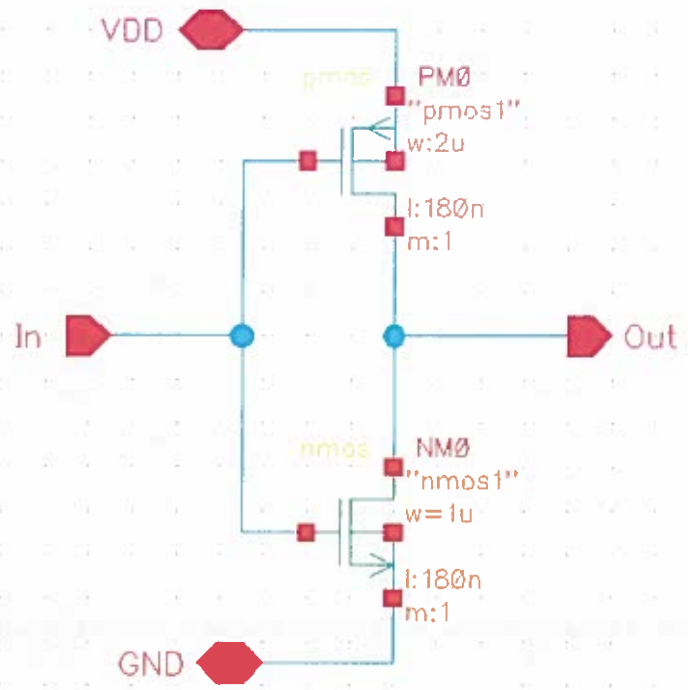


Figure 1.7: Schematic of an inverter



Figure 1.8: Symbol Generation

rectangle but you may want to make it more informative (Fig. 1.9).

Click on save symbol at the top of the toolbar. Close the symbol editor and the schematic of the inverter.

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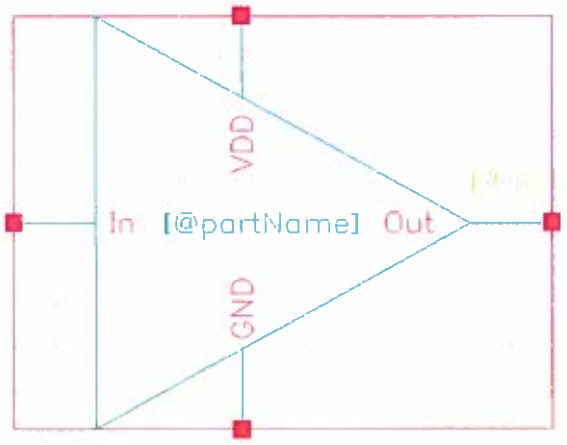


Figure 1.9: Changing the symbol design to that of a standard inverter symbol

1.4 Simulation using the analogue environment (Spectre)

1.4.1 Create a testbench

To simulate the inverter we just created, a testbench needs to be created as follows in the same library as you created your inverter:

1. create a new cell view and name it "inverter_tb"
2. add the inverter just created similarly to adding a NMOS
3. add vdd, gnd, vdc and vpulse from analogLib, and mimcap[2] from gpdk180
4. connect the instances as show in Fig. 1.10.
5. edit the DC voltage of vdc to 1.8
6. edit vpulse so that the DC voltage is dcin, voltage 1 is 0 V, voltage 2 is 1.8 V, delay time and pulse width are 500 ns, period is 1μs, rise time and fall time are 1 ns.
7. keep the default properties for mimcap.

The vdd and gnd instances you just added are inexplicit global pins and any nets connected to them are automatically assigned the name of "vdd!" and "gnd!" respectively. The exclamation mark means the label is global, i.e connected within all levels of the schematic. Now we need to create labels on the wires using create wire label button (Fig. 1.6, shortcut "l"). Then we add 3 labels (vdd!, In and Out) as shown in the schematic (Fig. 1.10). Note that all net labels are **CASE SENSITIVE**.

[2]MIMCAP stands for metal-insulator-metal (mim) capacitor

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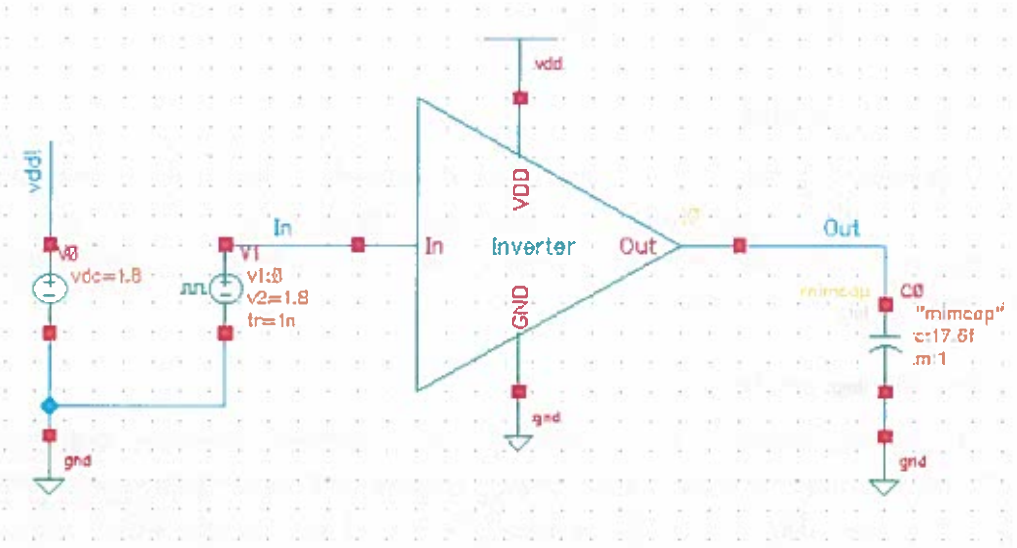


Figure 1.10: Schematic of inverter testbench

1.4.2 Start Analog Design Environment XL

To start the analogue environment, within the schematic editor click on the Launch menu and select ADE XL. Then opt for Create New View. After clicking several OKs, you will see Fig. 1.11. If there is a pop-up for using the ADE GXL license just choose "yes" or "always".

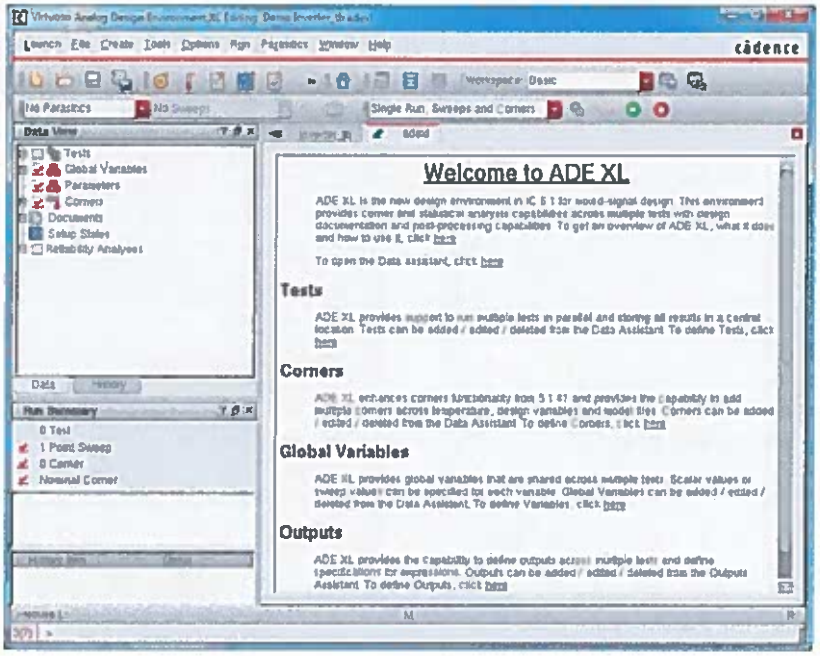


Figure 1.11: ADE XL window

1.4.3 Create a test

To create a new test, expand the Tests node in the Data View panel on the left hand side. Then single click click to add test and carry on with the default options in the pop-up window. After

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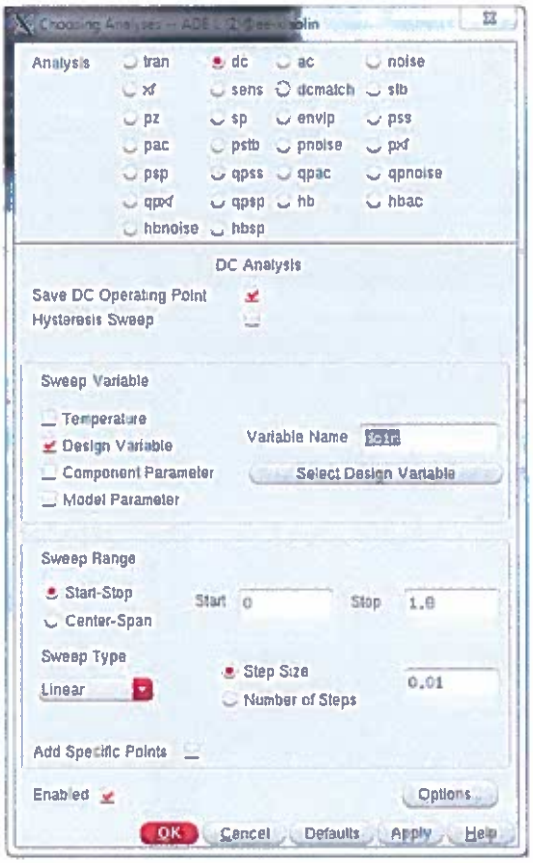
this, you should see the ADE XL Test Editor.

1.4.4 Design variables

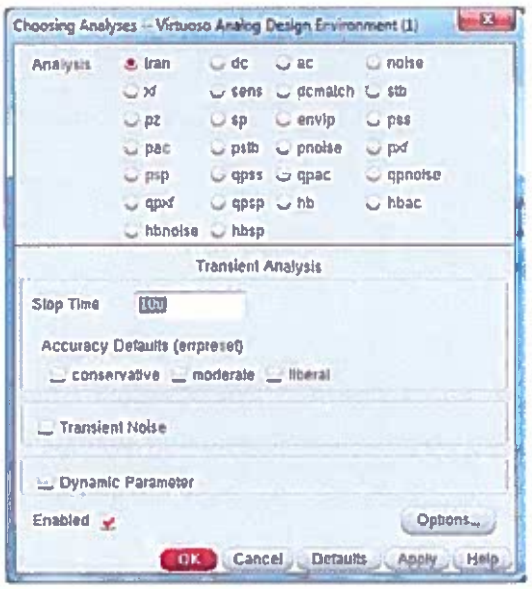
The variables created in the schematic need to be recognised by the simulation environment. To do this click on the Variables menu and select copy from cell view. You should now see the variable dcin in the left hand pane of the window. Set its value to 0. You must initialise all variables before you run simulations.

1.4.5 Setting up analyses

You can choose various analyses by opening the analyses dialogue. From the Analyses menu select Choose.... From this window select the dc analysis and set it up to sweep the design variable. Then select dcin, and set Sweep Range between 0 and 1.8 volts with a step size of 100 μV, as shown in Fig. 1.12a. Then setup a 10 μs transient analysis as shown in Fig. 1.12b.



(a) DC Analysis setup



(b) Transient Analysis setup

1.4.6 Setup Outputs

Open the Outputs menu and select To Be Plotted⇒Select On Schematic. The schematic will appear as the foremost window and here you can select the node voltages and currents which you wish to be plotted. There is a convention here: if you select a terminal, a coloured ring will appear indicating that the current into that node has been selected; If you select a net, the wire

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will be highlighted indicating that the voltage on that net is selected. Now, setup two outputs. One is the Out wire from the schematic and another one from the calculator as shown below.

Using Calculator

Open calculator from Tools⇒Calculator in the ADE XL Test Editor. Select the button vs. This button is used for plotting the results of a swept DC analysis. The other buttons present are for other types of analyses:

- vt/it–voltage/current in transient analysis
- vf/if–voltage/current in AC analysis
- vdc/idc–voltage/current in DC analysis
- vs/is–voltage/current in DC SWEEP analysis

You will then be asked to select the output voltage net ("/Out") of your circuit from the schematic. Notice that when you do this the expression in the calculator is updated. Then select cross (report the value of the sweeping parameter when the selected signal cross a set threshold) from Function Panel and set Threshold Voltage to 0.9 as shown in Fig. 1.13. The final look of ADE XL is shown in Fig. 1.14.

1.4.7 Simulation

Before running the simulation, make sure that you check and save all the schematics. Then, click the green play button on the toolbar. If you are asked to add your outputs to the saved list, click "yes". This usually happens when you want to save terminal currents.

When carrying out more complex simulations you may want to save the state of a simulation (including results). This saves the time needed to set up the analyses each time you re-open the circuit. To do this, right click the test name in the data view panel and select Save state. Enter a file name in the Save As box and click OK. You can reload a state by opening the Session menu and selecting Load state. You can use the save button to save your ADE XL setup. However, save state and save ADE XL setup are different.

1.4.8 Exploring the waveform

If you have completed this correctly, you can open the plots from the results tab by clicking the small graph icon showing the outputs. Now, you can do some simple measurements as shown in Fig 1.15. Here are some shortcuts:

- v – add a vertical line
- h – add a horizontal line
- m – create a marker
- d – create a delta marker (using the previous marker as a reference)
- Ctrl+e – delete all markers

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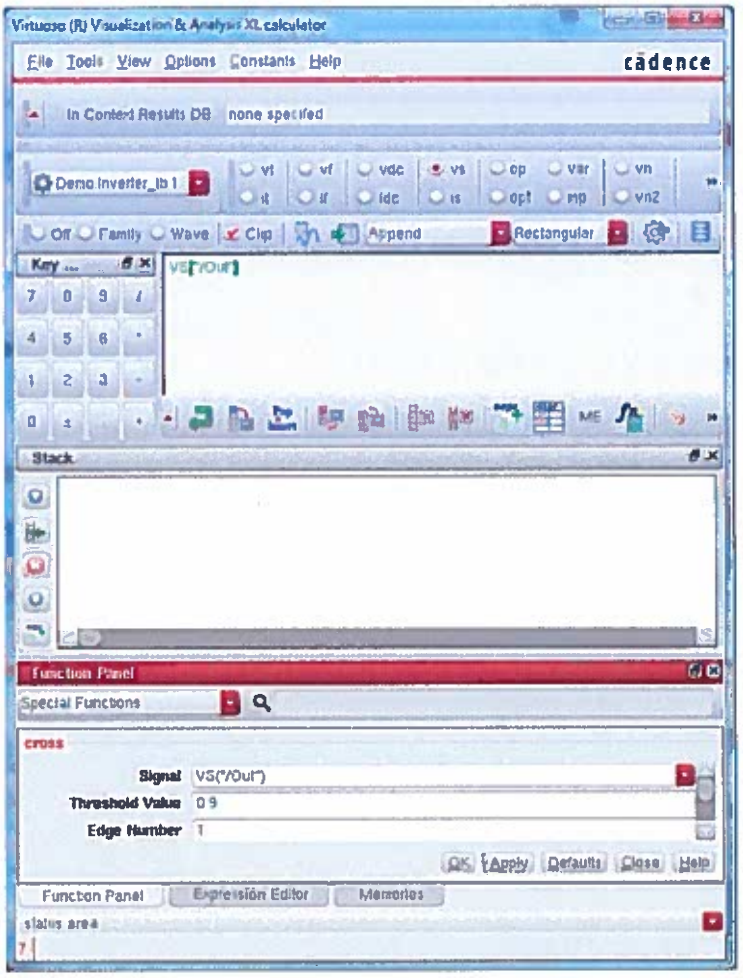


Figure 1.13: Calculator setup for measure switching point.

1.4.9 Printing results for your logbook

To print the schematic, click File and then Export Image and you will see the window shown in Fig. 1.16. Change the background to white, enter a file name and click Export.

To print waveforms (simulated plots) choose File⇒Print. Then print to PDF file.

1.5 Exercise

Create a a ring oscillator(Page 339 [1]), with no more than five stages, in a new cell called RingOsc. It should drive a load of 500 fF at the output (mimcap). The oscillation frequency should be 10±0.1 MHz at 27 degree Celsius. This oscillator will be used for the project.

Add (1) Schematic (2) output waveform (3) oscillation freq into your logbook.

Hints:

- To start oscillation, ramp up the power supply from 0 to 1.8 V with a rising time of 100 ns.
- The load needs an output buffer to be driven.(Example 3.10 Page 74 [1])

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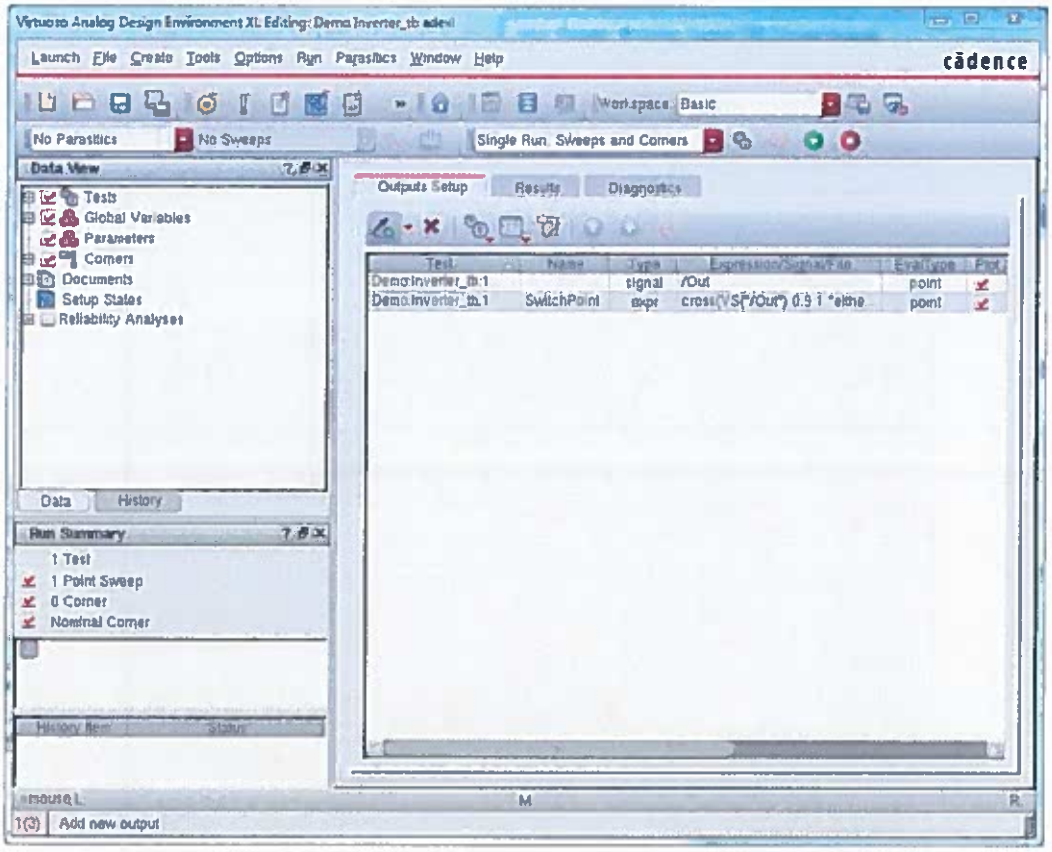


Figure 1.14: ADE XL window after setup

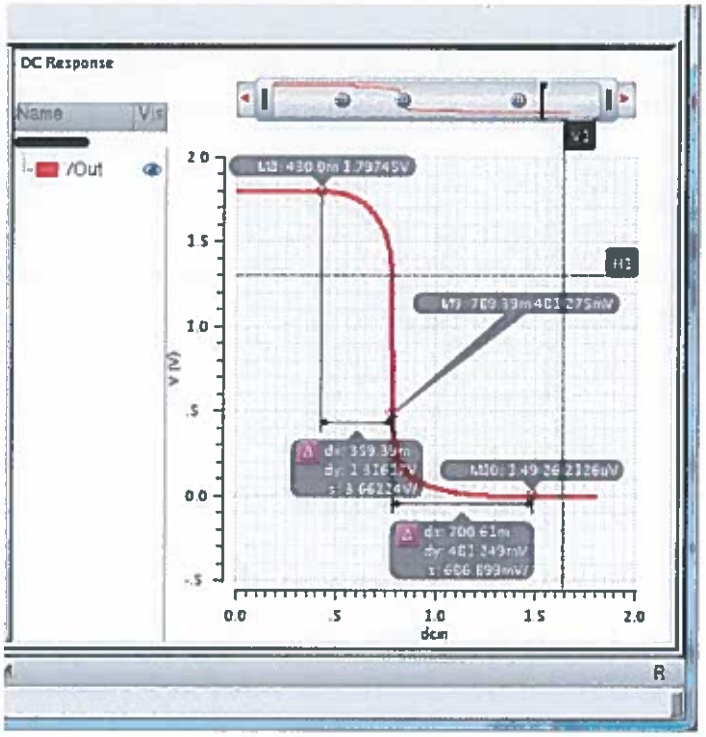


Figure 1.15: Types of available markers. This is for illustration only. There is no need to place the markers as above unless you require to annotate certain features.

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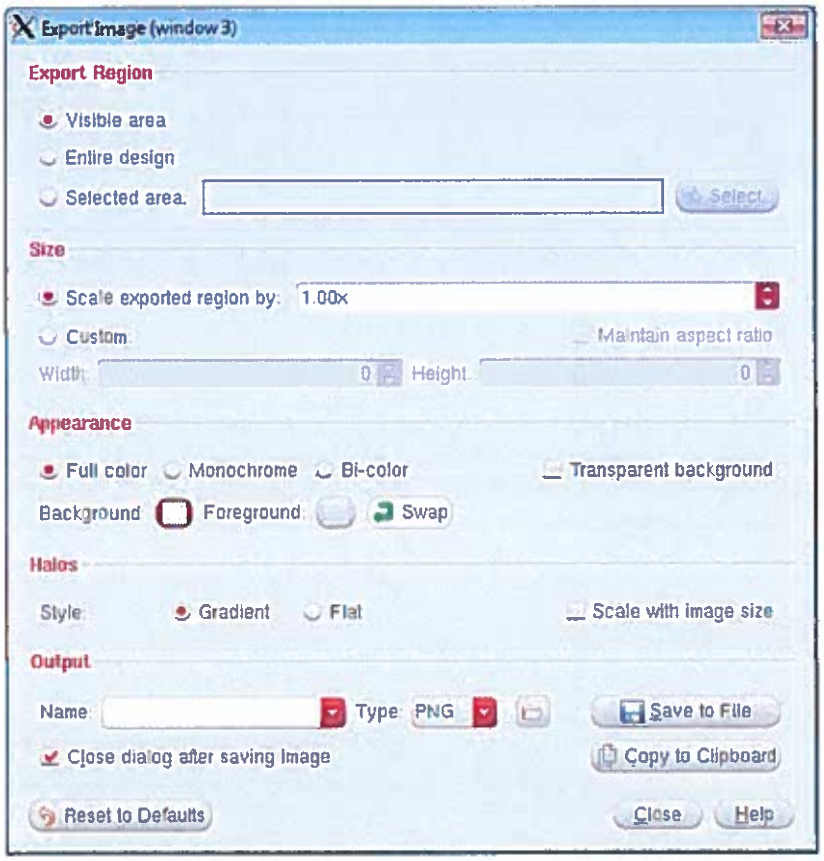


Figure 1.16: Save schematic with white background

- Parametric analysis helps. You can set inverter sizes by using global variables.
- Setting up specifications (in the output setup tab) may ease your task.

1.6 References

[1] R.Jacob Baker, *CMOS Circuit Design, Layout, and Simulation, Third Edition*. Hoboken, N.J: John Wiley & Sons, Inc., 2010, pp. 332–340

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Laboratory 2 Inverter Layout & Digital Cells

What you will learn

- How to use the Virtuoso Layout Tool.
- How to perform DRC, LVS and Extraction of your layout.
- Create a digital standard cell library including an Inverter, NAND gate and DFF with a height of 8 μm

2.1 Introducing Virtuoso Layout

The Virtuoso layout tool is polygon based. If two shapes from the same layer overlap, they are connected. The basic shape unit is μm. However, the design canvas is grid based, which is the minimum resolution a specific process can achieved. For 0.18 μm, the grid is 0.01 μm. If the layout is off-grid, it will fail the DRC and be rejected by the foundry. **Always ensure the grid is correctly configured when starting design.**

2.2 Create a layout of an inverter

In this lab, we are going to create the layout for the inverter from the previous lab. The target height is 8 μm.

2.2.1 Create the layout view

Open the schematic of the inverter and select Launch⇒Layout XL. Then create a new layout with default configurations. Then you will see a window as shown in Fig. 2.1. Click okay and you will see Fig. 2.2. On the top of the left hand side is the layer panel. Here, AV means All visible, NV – None visible (except the current selected layer), AS – All selectable, NS – None selectable (including the selected layer). You can toggle selectability and visibility for each layer using the check box next to the layer name. Most of the time, we draw shapes using the layers whose purpose is drawing.

2.2.2 Set grid and display options

In the main window select Options⇒Display and set the Display Stop value to 32. Pressing "SHIFT+F" in the main layout window has the same effect. Also, **changing X Snap Spacing**

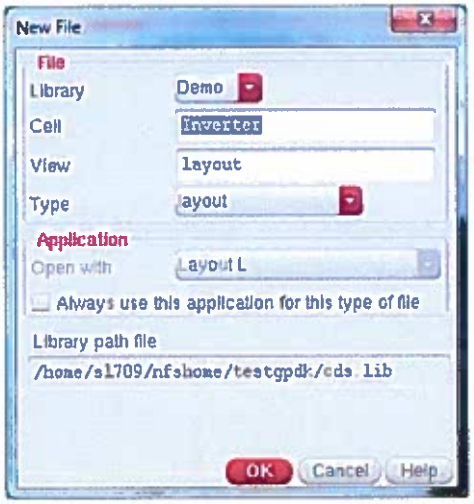


Figure 2.1: Create a layout view

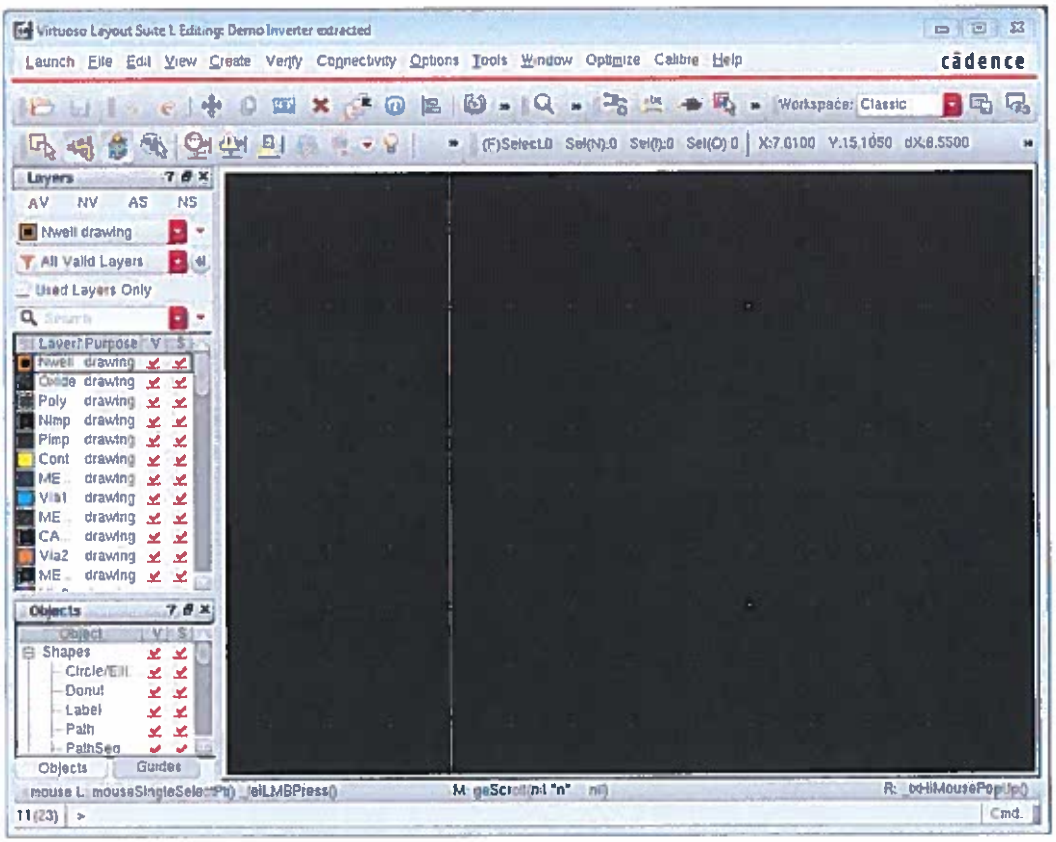


Figure 2.2: Virtuoso layout window

and Y Snap Spacing to 0.01. In addition, check the Dot Pins and Pin Names. This setting is shown in Fig. 2.3.

2.2.3 Start layout

To help you navigate the design, drag with right mouse button to zoom and "shift+z" to zoom out.

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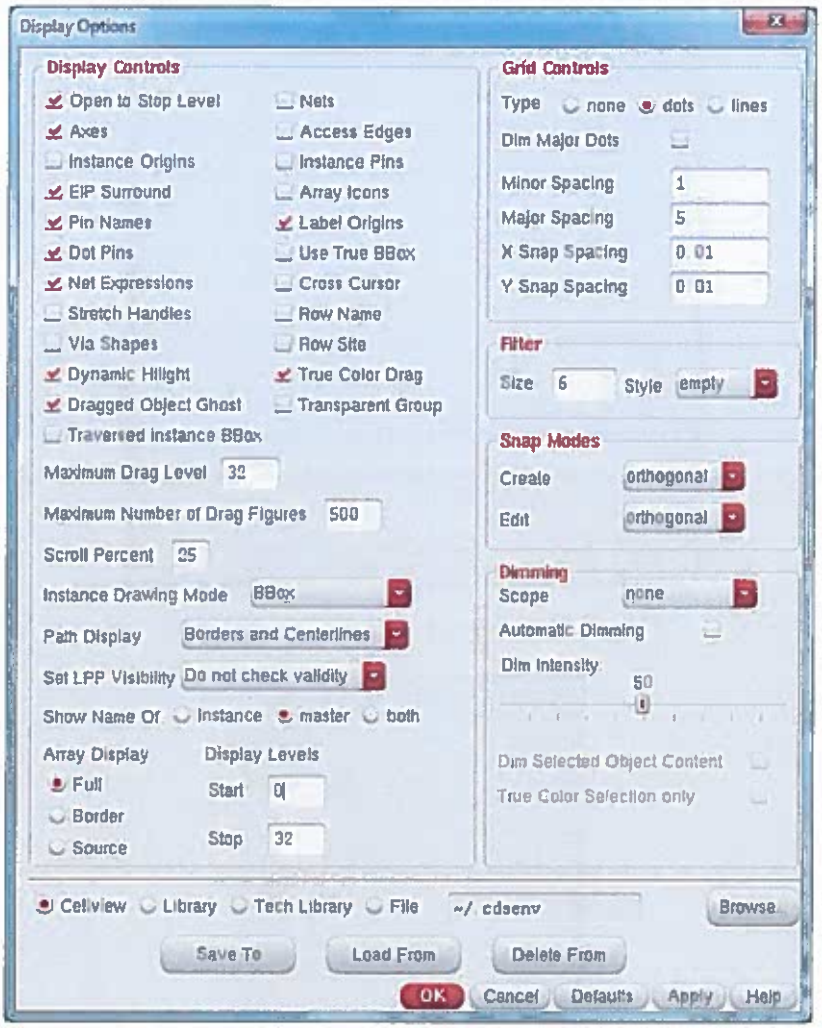


Figure 2.3: Display settings

Generate layout from schematic

Select Connectivity⇒Generate⇒All From Source to bring up the window shown in Fig. 2.4. The only parameters that need to be changed are in the I/O Pins tab. You should see a list of pins in the centre of the tab. Check Create label as Label and select Options in that tab. Then modify the options as shown in Fig. 2.5. Then you can proceed. If you want to add individual devices directly into the layout, you can do the same thing as in the schematic; just press "i".

Splitting gate into multiple fingers

For a wide gate, it would be beneficial to split into multiple fingers to keep the FET aspect ratio down thus easier to meet the height requirement for standard cells. Here, we need to split the PMOS gate into two fingers. Edit the properties of the PMOS by selecting it and press "q". Switch to parameter tab and enter 2 in fingers. To create a body connection, edit the properties of the NMOS and select bodytie type as detached

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Figure 2.4: Generate layout from schematic



Figure 2.5: Set pin label properties

Annotation Browser and DRD

One useful assistant is the annotation browser which can show any incomplete nets. This can be considered like an online layout vs schematic (LVS) check. You can invoke it from Windows⇒Annotation Browser. You can also turn on the DRD tool. It performs real time Design Rule Checking (DRC) which we will investigate in depth later. In brief though, the DRD tool runs a continuous DRC and highlights any rule violations immediately as you are laying out the circuit. The likelihood of there being design rule violations when you run the full DRC later is thus much reduced. To turn on the DRD go to Options⇒DRD Edit and change interactive mode to notify and check post-edit checking. **However, this is NOT to be used as a substitute for the final DRC as this does NOT implement the full set of rules.**

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Create Via – Connection between layers

Generally, it is not advisable to use poly to make long connections. Therefore we need to create a poly to metal 1 via for the gate and make connection there. To do so, select Create⇒Via (shortcut "o"). The via creation form is shown in Fig. 2.6. Be sure to use the correct type of via:

- M1_POLY1 – Poly to metal 1
- MX_MY – Metal X to metal Y

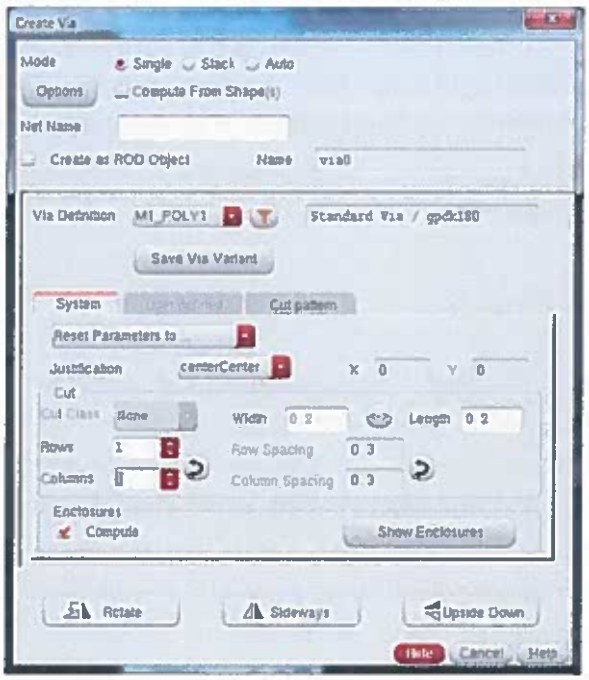


Figure 2.6: Creating a via from poly to Metal 1

Create guard rings and body connection

Guard rings connect metal layers to the substrate or doping layer and are used to prevent the injection of minority carriers from one type of transistor (say NMOS) into the other (say PMOS) where they become majority carriers and interfere with the correct operation of the transistor. They are also used to create the substrate contacts required to form a complete transistor. You need two guard rings in your design. PMOS placed in an N-well needs an Nguardring tied to the highest potential in the design (usually vdd). NMOS needs a Pguardring tied to the lowest potential (i.e. the substrate).

However, for digital standard cells, guard rings are usually not a full circle. To create a guard ring for PMOS, select Create⇒Multipart Path then press "F3"and select Pguardring in MPP template. Then you can draw the path. For the PMOS, select Nguardring. Once you have drawn your PMOS guard ring draw a rectangle (shortcut "r") of Nwell around its guard ring (Create⇒Shape⇒Rectangle). This is compulsory to ensure that the guard ring and all the PMOS transistors are in the same well.

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Making Connections

Once you've added all of your transistors you need to add metal paths to connect the terminals together so as to match the schematic. To create a wire (or path), you can use Create⇒Shape⇒Path or the shortcut "p". Make the inverter input and output connections on metal 2 by selecting this layer before starting to draw the wire. Metal 2 is used here so that the wire goes over the guard ring. Do not forget to create the via between metal 1 and 2.

About pins

The pin comprises a pin shape and a label. It only works when the central cross of the label is on the shape (label is attached to the pin). Also, the shape and the label have to be on the same layer. Note that we created all of our pins on metal 1 as standard cells must only use metal 1 and poly. Then drag all the pin shapes (the label should move with it if properly attached) to the correct location. The final look is shown in Fig. 2.7.

2.3 Design Rule Check

Not everything that can be drawn in the layout editor can be correctly fabricated. Therefore, the foundry has supplied a set of design rules for designers to follow. Checking your design against these rules is known as Design Rule Checking (DRC) and you should run DRC regularly to avoid errors accumulating. **Your final design must pass the DRC.**

To check your design, select Verify⇒DRC. You will be presented with a dialogue as shown Fig. 2.8. Just leave everything by default and press OK to run the DRC check. If some fields are missing, fill them manually according to Fig. 2.8.

When the run is complete the results will be displayed in the CIW window. If there is an error (like in Fig. 2.9), select Verify⇒Marker⇒Find. This will bring a tool to help you locate the errors.

Note: You can ignore any DRC results about Ratio Coverage and INFO. The most important DRC errors are spacing, minimum width and contacts.

2.4 Layout vs Schematic

Before completing you must ensure that what you have drawn exactly corresponds to what you wanted in your schematic. A tool known as Layout Versus Schematic (LVS) automates this process. It is very important and should always be completed. To run the LVS, we need to first extract the netlist from the layout. Select Verify⇒Extract. The form is shown in Fig. 2.10. Just click "OK". Again, if any fields missing, you need to complete them manually.

Then open the extracted view and select Verify⇒LVS(Fig. 2.11). The results will be displayed in a message. If there is an error, select Verify⇒Marker⇒Find and you will see the debug tool (Fig. 2.12).

Note: Attach the final layout as well as proof of a DRC and LVS clean layout in the logbook.

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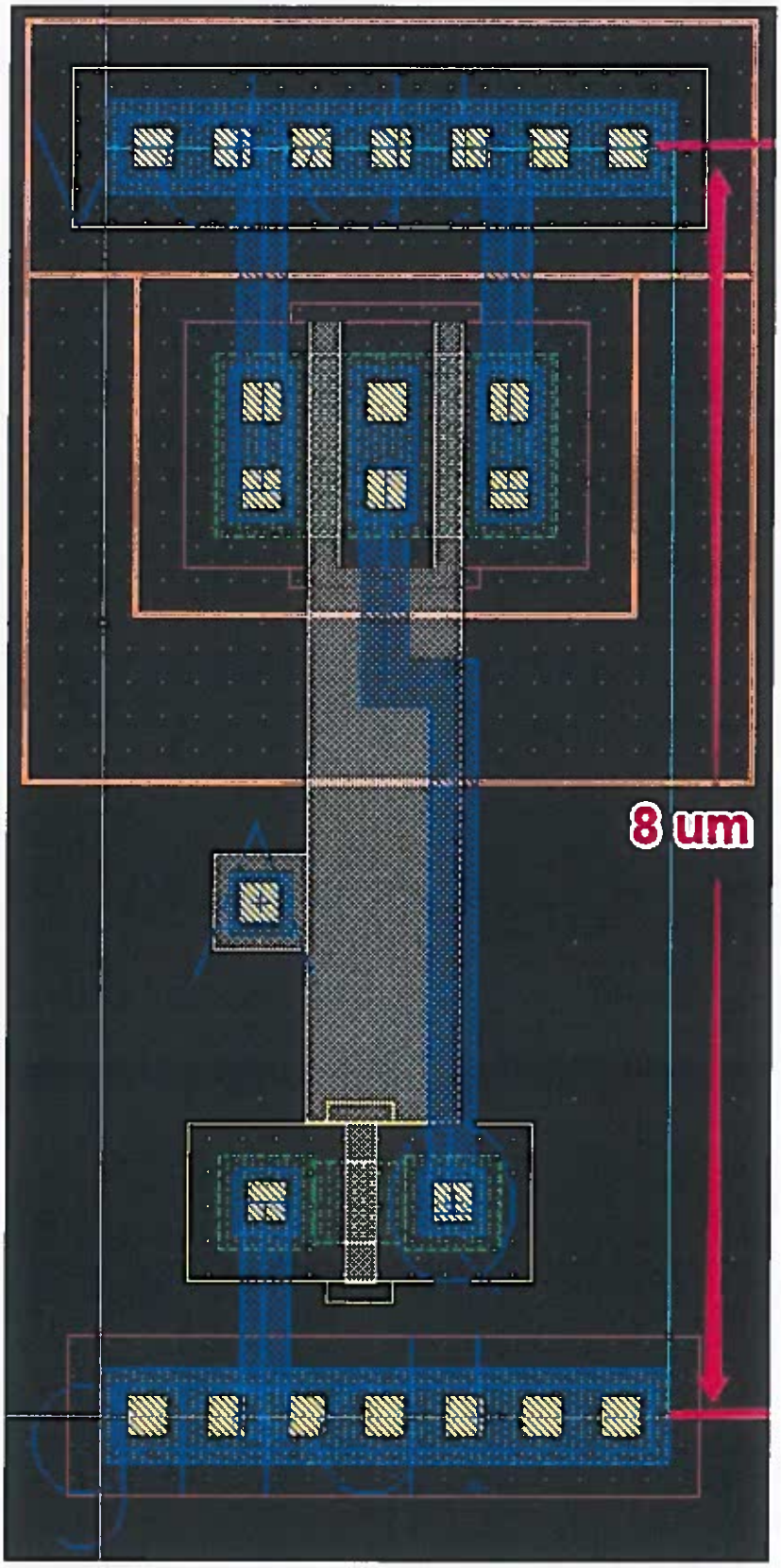


Figure 2.7: Final look of the layout. The cell height is 8 μm.

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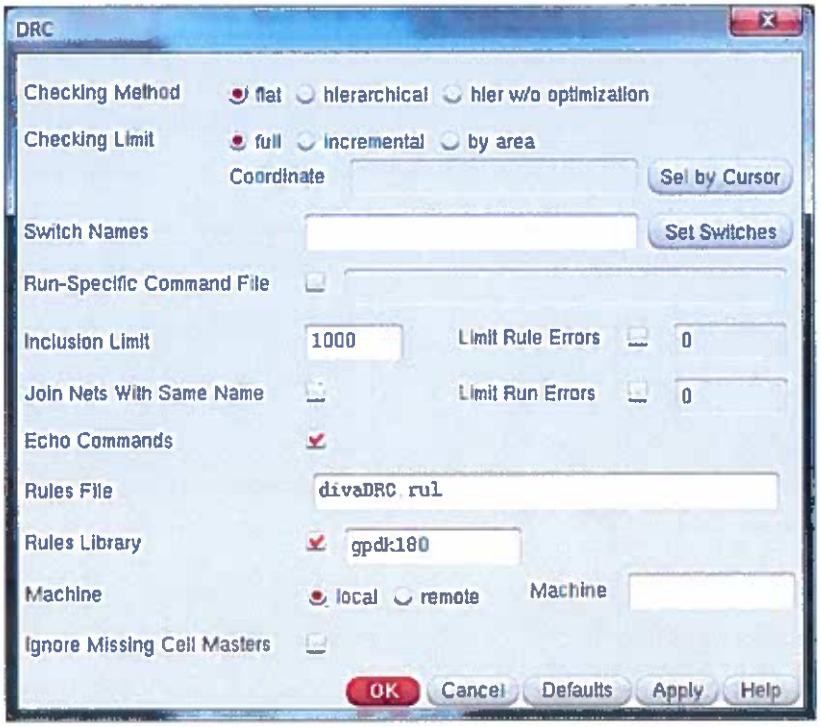


Figure 2.8: DRC dialogue

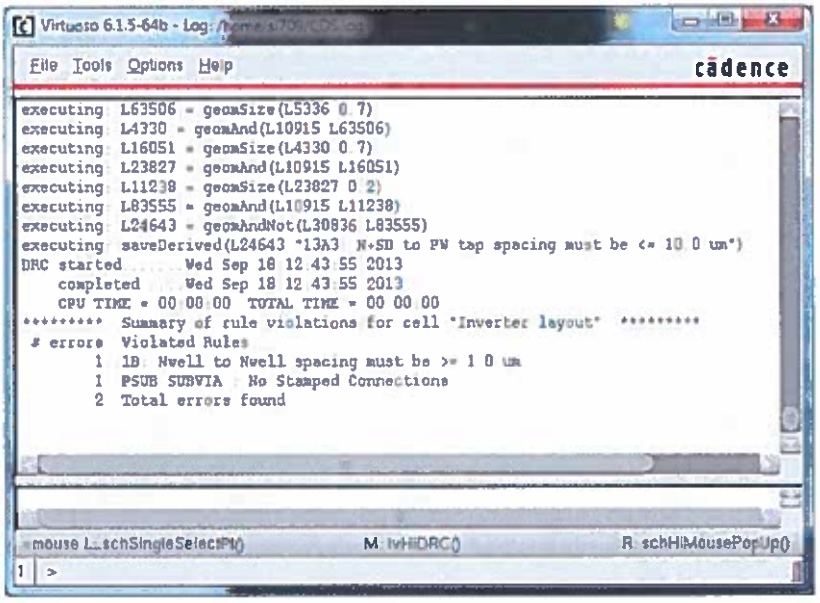


Figure 2.9: DRC error message in CIW.

2.5 NAND Gate

You are required to design a NAND gate with a cell height exactly **8 μm**. Following the steps in Lab 1, create a cell called "NAND" and produce the schematic for the NAND gate in it. You can generate a "Cellview" as described in Lab 1. Also, create its testbench "NAND_tb" and place the NAND cell in it (the same way you add an instance). Here you will simulate your design of the gate using DC and Transient simulations.

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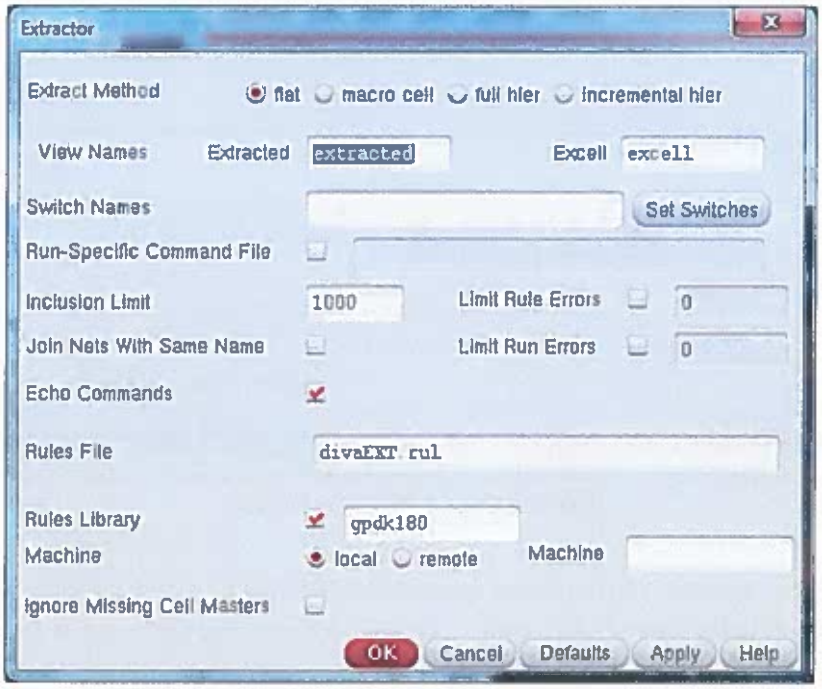


Figure 2.10: Extract dialogue.

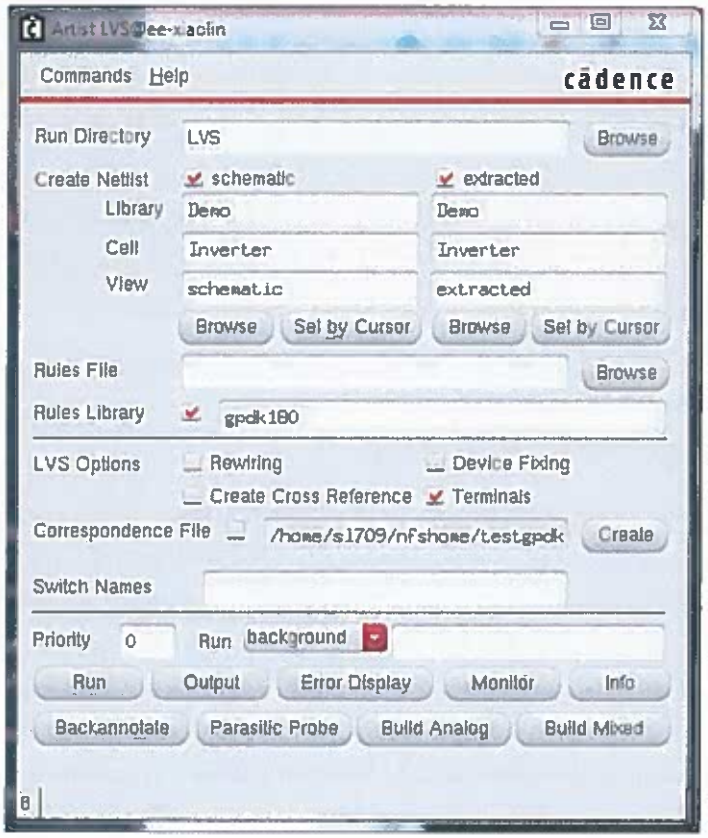


Figure 2.11: LVS dialogue.

The load is a 1 pF mimcap capacitor from library gpdk180. The NAND gate should be balanced for best noise immunity, i.e. the switching point (when both inputs are changing) is

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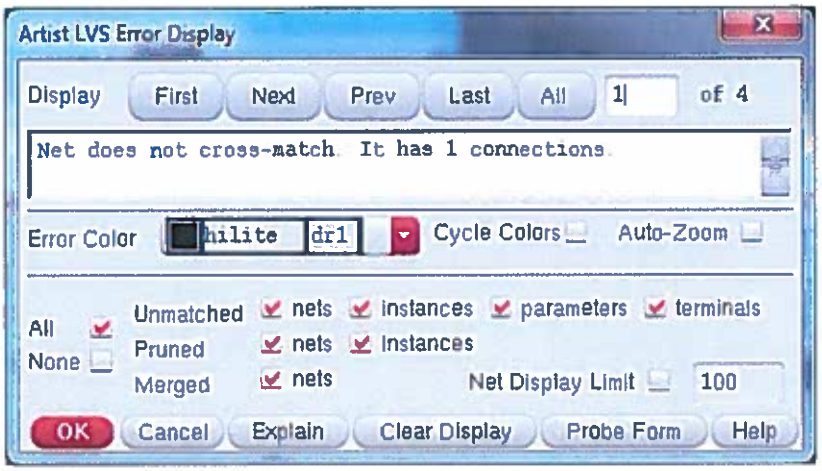


Figure 2.12: LVS Error Display

at half the supply. You can verify this by doing a DC Sweep simulation as shown in Lab 1. The worst propagation delay (for NAND gate, you need to specify which one of the two inputs is changing) must not exceed 10 ns. Verify this using a transient simulation as shown in Lab 1. When you finish your gate schematic design and simulation, proceed by generating the NAND gate layout.

(1) Attach schematics, layout and simulation results in the logbook; (2)Attach proof of a DRC and LVS clean layout
Hints:

- Try to achieve minimum propagation delay while keeping the layout size (footprint area) to a minimum. Attach transistor sizing in the logbook and reasoning.
- You can find examples of compact NAND gate layout online.
- You will to use this cell for your project.

2.6 Edge triggered D Flip-Flop

The edge triggered D Flip-Flop is shown in Fig. 2.13 (without transistors sizes). (If you are unfamiliar with this circuit, refer to the references provided at the end.) This operates at 100 MHz. Reproduce the schematic in your library with a cell named "DFF". Also, create its testbench "DFF_tb" as shown in Fig. 2.14. The load is a 4 μm by 4 μm mimcap from gpdk180.

Run a simulation with the testbench to show that the DFF is working. A sample result is shown in Fig. 2.15 for a 100 ns transient simulation. It can be seen that the DFF changes its output at the rising edge of the clock signal. You don't need to reproduce the exact waveform, but need to show that everything works. After you finish with the simulation, proceed with generating the layout of the DFF. The cell height must be exactly **8 μm**

(1) Attach schematics with transistor sizing, layout and simulation results in the logbook; (2)Attach proof of a DRC and LVS clean layout

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Hints:

- Aim for a compact layout. You can find examples of compact NAND gate layout online
- You will to use this cell for your project

2.7 References

[1] R.Jacob Baker, *CMOS Circuit Design, Layout, and Simulation, Third Edition.* Hoboken, NJ: John Wiley & Sons, Inc., 2010, pp. 375–389

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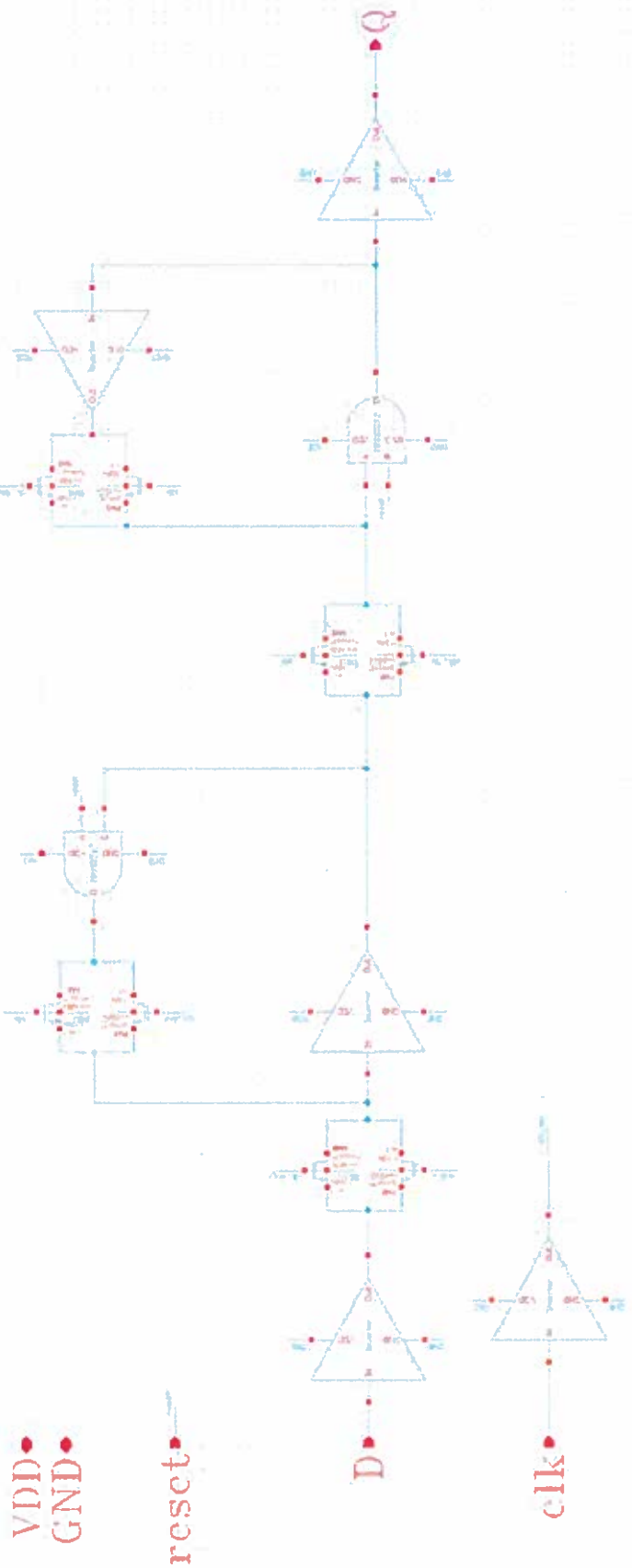


Figure 2.13: DFF schematic

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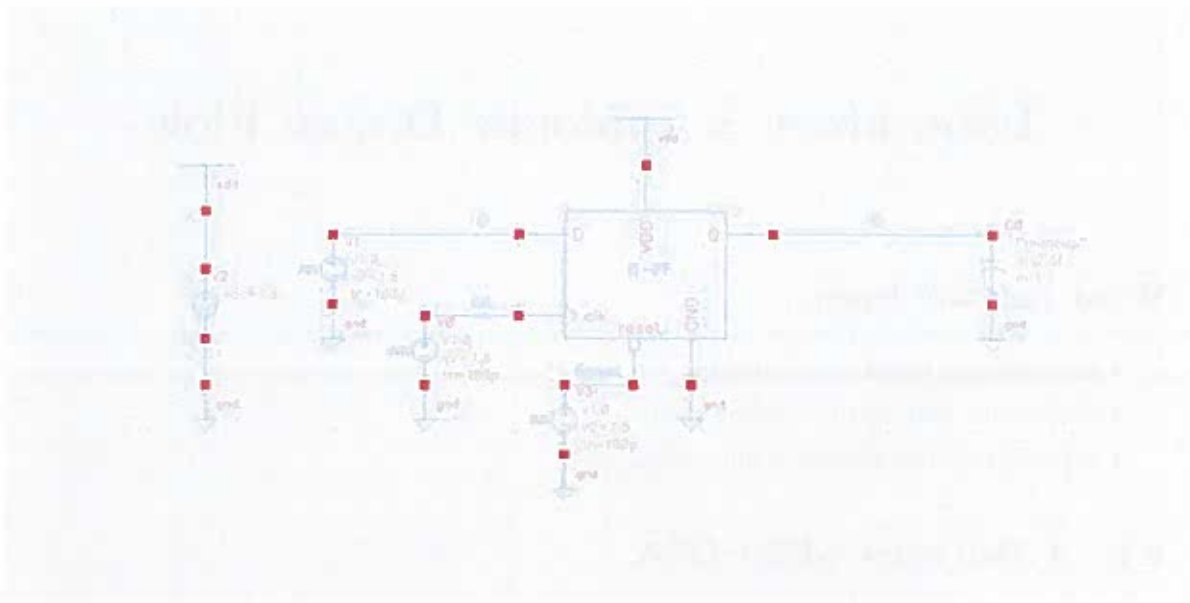


Figure 2.14: Testbench Schematic

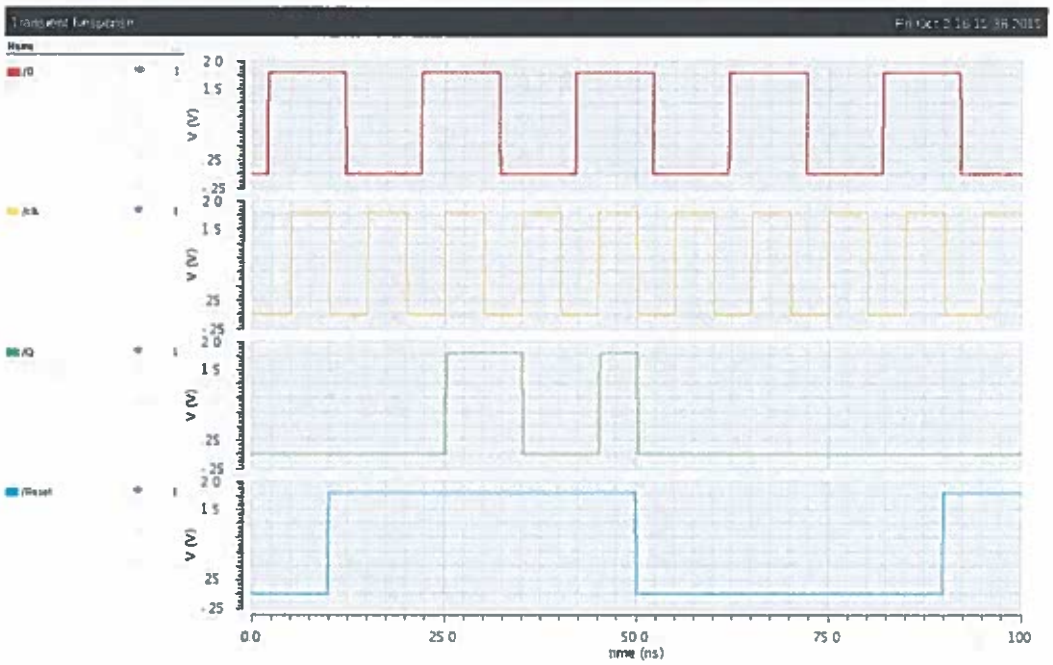


Figure 2.15: Simulation results of the DFF

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Laboratory 3 Analogue Design Flow

What you will learn

- Designing an operational amplifier
- Performing amplifier characterization
- Introduction to analogue layout techniques

3.1 A Two stage Miller OTA

The two stage miller OTA is a commonly used in analogue circuit design [1–3]. For example, it may be used as a buffer. Create a cell called opamp as shown in Fig. 3.1. The current mirror **PM2-PM5** allows easy biasing of the differential pair **PM0-PM1** with the current generated via **R0** and **PM5**. The NMOS transistors **NM0-NM1** are active loads for the differential pair, folding transductive current generated by a differential input signal **INP** & **INN** towards the output. An important characteristic of this topology is miller compensation capacitor **C2** which splits the two poles for a stable close loop operation.

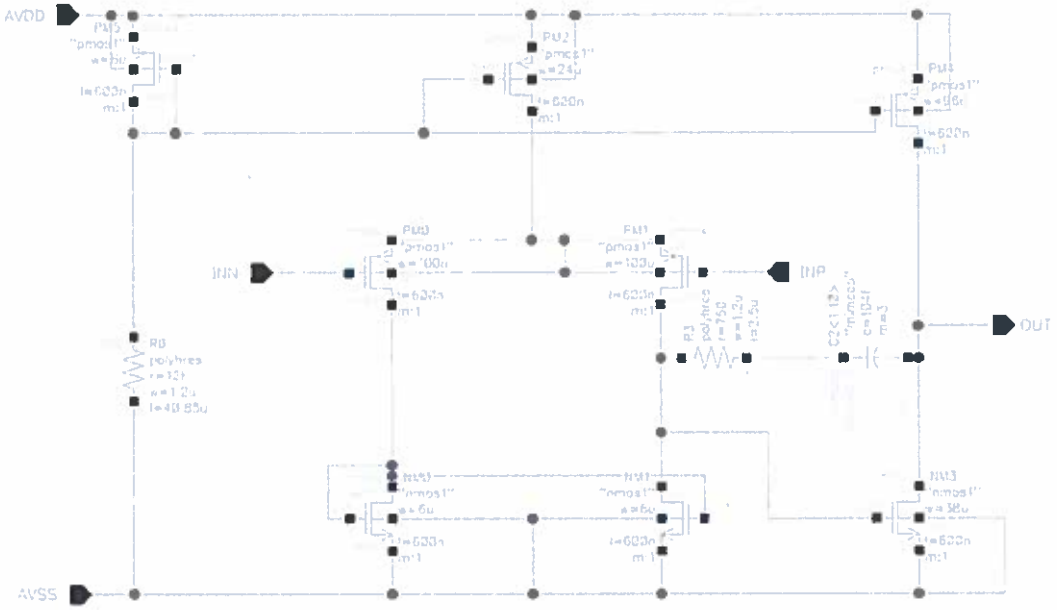


Figure 3.1: A two stage miller OTA

Create a new cell called opamp_tb as shown in Fig. 3.2. Power up the circuit with a DC voltage source vdc and a Voltage Controlled Voltage Source vcvs with gain of −1. Here **V1**

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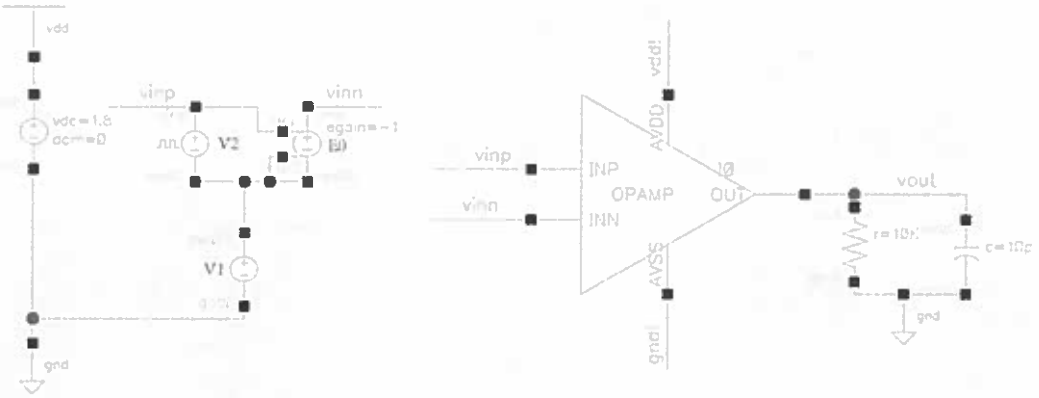


Figure 3.2: Testbench

provides the common mode input signal and **V2** & **E0** introduce the differential input voltage. The properties of each source are summarised below. Load resistor and capacitor are from analogLib.

- V0 – VDC source for the supply (dc = vdd)
- V1 – VDC source for the input common mode (dc = vcm)
- V2 – VPULSE source (dc = 0, ac = 0.5, v1=vcm, v2=-vcm, period=5u, pulse width=2.5u)
- E0 – VCVS source with gain -1

n.b. Ensure that transistor sizing and interconnect is equivalent to that in Fig. 3.1 before we start characterization.

3.2 Amplifier Characterization

This characterization procedure starts by using the simulation tool for DC and small signal analysis. After starting ADE XL introduce the variables used for some of the voltage sources Variables ⇒ Edit and define the variables vdd = 1.8, vcm = vdd/2. Similarly set up the tabs as shown in Fig. 3.3. We will be using DC analysis to sweep the differential input to observe both the output swing and operation of the transistors for certain operating conditions.

The AC analysis calculates the small signal transfer function of the circuit. Notice that the AC differential magnitude at the input of the amplifier is 0.5 (so the equivalent input magnitude is 1) such that the output signal gain is normalized appropriately. Run the simulation and browse the results with Tools ⇒ Results Browser which should automatically load the resulting psf file. Expanding some of the trees in the browser section lets you navigate to the 'dc' tab. right click and plot some of the signals to see how the varying input affects the voltages on different nodes. Then navigate to the 'ac' tab and plot vout. Change the y-scale so that it is in dB by right clicking on the waveform and selecting Dependent Modifier ⇒ dB20. Similarly you can right click the waveform and select Send to ⇒ Calculator to evaluate expressions such as phase(VF('\ vout')) or dB20(VF('\ vout')). Finally check the 'tran' tab to check slew rates associated with the square wave input when actively driving the load. Using Results ⇒ Annotate ⇒ DC Operating Points, Confirm all transistors are in saturation by checking their operating conditions.

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Figure 3.3: Operational amplifier characterization setup

3.3 Analogue Layout

In order to assure the functionality and quality in analogue design, making sure transistors exhibit similar behaviour even after process variation is essential. These techniques allow us to prevent imperfections such of offset, instability, and possibly latch-up. The designer needs to identify which specific groups of transistors need to have very similar characteristics such as IV behaviour. This includes differential pairs and current mirrors where it is important that the currents are exactly matched even if there is a gradient in the mobility of electrons across the wafer. The principle of such techniques lies with common centroid configurations. For an even higher requirement on matching, the introduction of dummy transistors further allow stress effects and short distance eching gradients to be more carefully matched.

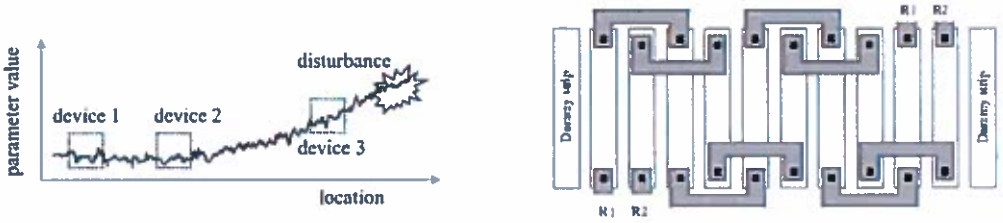


Figure 3.4: (left) illustration of the variation in parameter (i.e. oxide thickness) across a distance subject to imperfections or disturbances. (right) example of an inter-digitated resistor layout with dummy devices.

To start analogue layout, in your opamp schematic window (not the testbench one) click Launch ⇒ Layout XL to start the tool and choose a 'Create New' Layout with an 'Automatic' configuration and confirm the layout name. This allows us to automatically generate transistor primitives from the schematic. Import the primitives Connect ⇒ Generate ⇒ All From Source ⇒ Ok and let the tool do some pre device placement with Place ⇒ Analog ⇒ Automatic Placement ⇒ Ok.

Now you can rearrange some of the devices to improve matching and possibly make inter-

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connect easier. Remember to keep transistor alignment the same for important pairs. It is also useful to group all PMOS & NMOS transistors that have the same body connection so they can share the NWELL or guard ring. You can use the routing assistant by pressing 'ctrl+shift+W' while making connections and 'space' to introduce vias along the way. The assistant will highlight the connections that need to be made with the initiated net. At this point try to break up transistors to put them into common centroid configurations. While making changes to the schematic the layout can actively be regenerated with Connect ⇒ Update ⇒ Components. Fig. 3.5 shows an example how a differential pair with size 4 μm/4 μm can be matched by using a 4 segment common centroid configuration with dummy transistors.

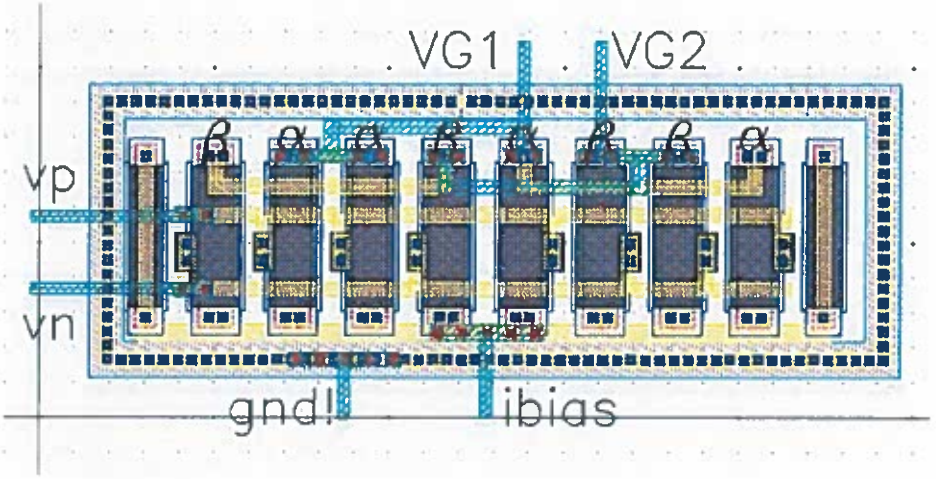


Figure 3.5: Example of a common centroid configuration BAABABBA with dummies on either side for a differential pair with size 4 μm/4 μm each.

3.4 Exercise

Add the AC, DC, and transient plots to your logbook and answer the following questions. (Ideally based on theoretical calculations. Keep Vdd fixed.)

- Schematics with sizes annotated (in case you have changed them for layout).
- Can you summarize Gain, 3dB cut-off frequency, and linear input range (is here defined as transconductance dropped by 1% from its maximum value) from operational parameters, e.g. power supply voltage, overdrive voltage, biasing current, W/L, etc?
- How can you improve gain with this topology?
- How does the slew rate relate to the biasing current?
- Layout with DRC/LVS results (print screen is accepted but must be readable).

3.5 References

[1] Behazd Razavi, *Design of Analog CMOS Integrated Circuits.* New York, NY: McGraw-Hill. 2001, Chapter 9

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[2] R.Jacob Baker, *CMOS Circuit Design, Layout, and Simulation, Third Edition.* Hoboken, NJ: John Wiley & Sons, Inc., 2010, Chapter 24

[3] E4.20 Lecture notes, *Lecture 8*

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Quick Reference Card

Basic *nix Commands

*nix command entry is CASE SENSITIVE
All commands are terminated with the return key or the ; (semicolon) character.

ls	List the contents of the current directory
ls -a	List the current directory, and include all files starting with dot characters
ls -al	Perform a detailed listing of the current directory
pwd	Report the current directory
cd *dir1*	Change to a sub-directory *dir1*
cd /home-/user1	Change directory using an absolute path */home/user1*
cd ~	Move to your home directory
cd ..	Move up a directory
mkdir *dir1*	Create a new directory called *dir1*
rm *file1*	Remove a file *file1*
rmdir *dir1*	Remove an empty directory called *dir1*
rm -rf *dir1*	Recursively remove directory *dir1* and all files without prompting. **WARNING: use this command with extreme caution**
cp *file1 file2*	Copy *file1* to *file2*
mv *file1 file2*	Rename *file1* to *file2*
less *file*	View a *file*
more *file*	Older command to view a *file*
./command	Run an executable or a script called *command* from current directory.
command	Run an executable or a script called *command*. This file has to exist somewhere in your path. (Otherwise you need to specify the full path)
source *scr.csh*	Run script *scr.csh* in the current C-shell
find *dir1*	Find all files in all directory within *dir1*
find -type d ./	Find all directories within current directory
grep *str file*	Search for string *str* in *file*
find | grep *zz*	Find files or directories containing the string *zz* in their names
who	List logged in users
whoami	Display your current username
tar cvzf *file.tar.gz dir*	Create a compress archive file *file.tar.gz* for directory *dir* in the current directory in verbose mode

Special *nix Characters

;	Command separator
&	Background execution
.	Current directory
..	Parent directory

*nix File Permissions

*nix file permissions are controlled by file access bits. If you list the files in a directory by typing "ls -l", you will see these on the left hand side of the report:

```
-rwxr--r--  1 user1 circ 34562 Sep 23  12:31 file1
drwx------  2 user1 circ  4096  Oct 8  15:56 dir1
lrwxrwxrwx  1 user1 circ    12  Oct 1  09:34 a -> b
```

These fields control the access to files, directories and symbol links. The bits work as follow:

	type	user	group	other
file	-	rwrx	rwx	rwx
directory	d	rwrx	rwx	rwx
symbol link	l	rwrx	rwx	rwx

Read permissions(r) – Allows read but not modify Write permissions(w) – Allows modification or deleting a file Execute permissions(x) – Allows you to run the file as a program

Start Cadence

Right click on the desktop and select terminal

- *cd nfshome*
- change to your working directory (or create one and change to it)
- *pdk gpdk*
- *cadence gpdk18*
- Design trade-offs (if any) and reasons for choosing a specific design
- Results from simulations (i.e. tables and graphs) with clear labelling (axis, units, etc.)

Cadence Schematic Shortcuts

Run the following command in a terminal.

```
find ~ -name \*cdslck | xargs rm
```

This should remove all the files that's been locked from last session.

Cadence Schematic Shortcuts

Check and save	shift−x
Insert instances	i
Create wire	w
Create pin	p Edit properties
q	
Change command options	F3
Zoom to fit	f
Zoom out	shift − z
Zoom	z
Loupe	.
Copy	c
Undo	u

Cadence Layout Shortcuts

Open display settings	e
Create via	o
Create route path	p
Create label	l
Draw rectangle using current layer	r
Copy	c
Undo	u
Startrouting assistant	ctrl−shift−x
Zoom to fit	f
Zoom out	shift − z
Zoom	z
Loupe	.
Create ruler	k
Clear all rulers	shift−k
Create guard ring	shift−x

Cadence Waveform Display Shortcuts

Add a vertical line	v
Add a horizontal line	h
Create a marker	m
Create a delta maker	d
Delete all markers	ctrl−e

EEE Cadence Servers

only available on campus or via college VPN

- ee-mill1.ee.ic.ac.uk
- ee-mill2.ee.ic.ac.uk

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